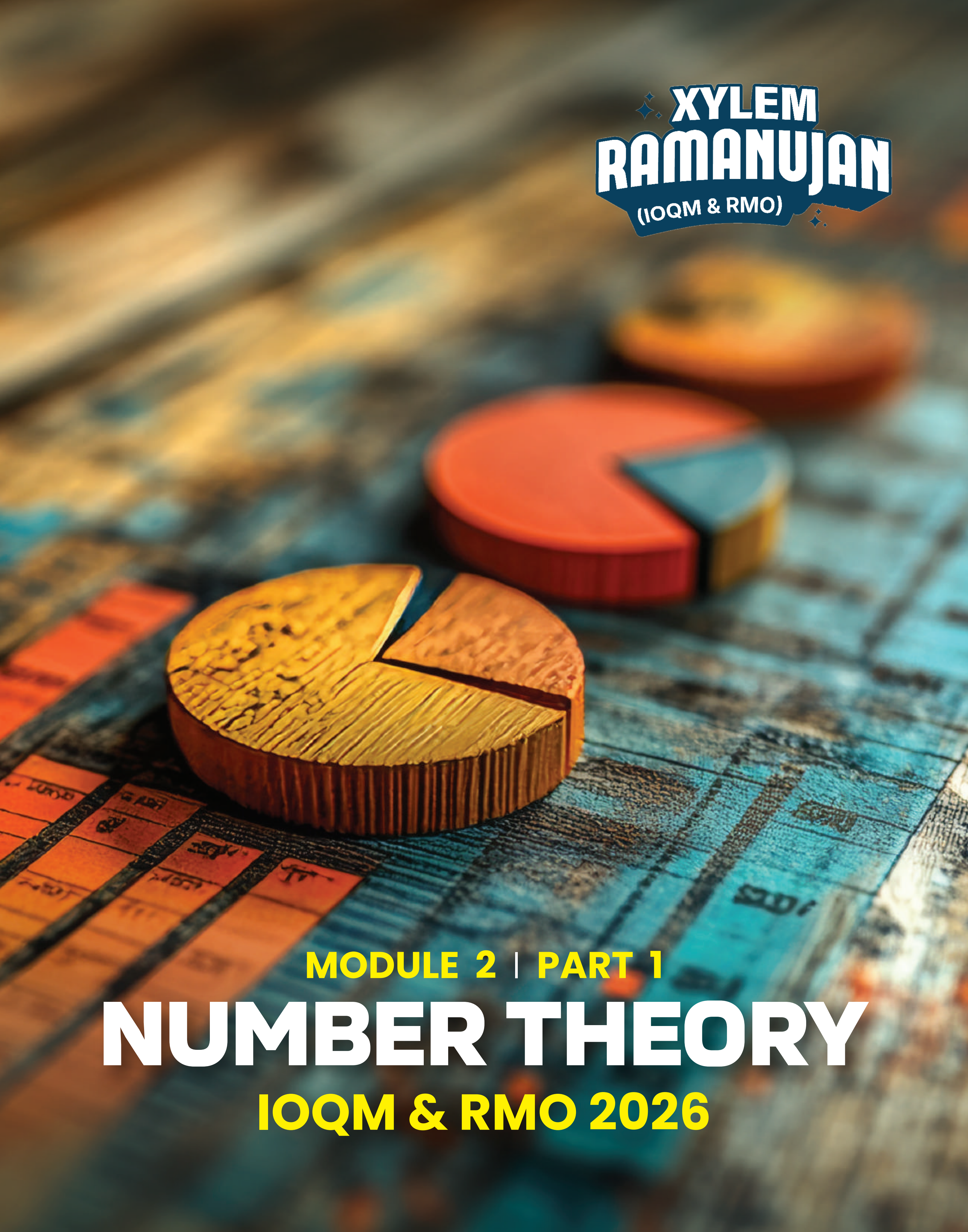

XYLEM RAMANUJAN (IOQM & RMO)

MODULE 2 | PART 1

NUMBER THEORY

IOQM & RMO 2026

NUMBER THEORY 1

Factors and Multiples

Factors

A factor is a number that divides the given number exactly, without any remainder. When a number n is divisible by b, we can say that b is a factor of n.
E.g., the factors of 16 are $1, 2, 4, 8$, and 16, because these numbers divide 16 exactly, without leaving any remainder.
In other words, 'a' is a factor of 'b' if there exists a relation such that $a \times n = b$, where 'n' is any natural number.

Properties of Factors

- 1 is a factor of any number.
- The greatest factor of any number is the number itself.
- A factor of a number is always less than or equal to the number.
- Factors of a number are finite and they lie between 1 and the number itself (both inclusive)

Multiples

A multiple of a number is the result you get when you multiply that number by any natural number $(1, 2, 3, ...)$.

E.g., multiples of 3 are 3, 6, 9, 12, 15, ...

(because $3 \times 1 = 3, 3 \times 2 = 6, 3 \times 3 = 9$, $3 \times 4 = 12$ and so on)
In other words, a is a multiple of b if there exists a relation of the type $b \times n = a$.

Properties of Multiples

- A number has infinitely many multiples.
- A multiple is always greater than or equal to the given number.
- Every number is a multiple of 1.
- If B is a multiple of A, then A is a factor of B.

Prime Numbers

A number which can only be divided by either 1 or itself is called a prime number.
Example: $2, 3, 5, 7$, etc.

Composite Numbers

A number which has more than two factors is called a composite number.

Example: $4, 6, 8, 10$, etc.

Example:

What are the factors of 9?

Solution:

The factors of 9 are the numbers which when multiplied together results in the number 9. Thus,

$1 \times 9 = 9$

$3 \times 3 = 9$

$9 \times 1 = 9$

Hence, the factors of 9 are $1, 3$ and 9

Example:

How many prime factors does 30 have?

Solution: (3)

As we know, the factors of 30 are

$1, 2, 3, 5, 6, 10, 15$ and 30.

The prime factors of 30 are $2, 3$ and 5.

Example:

What is the sum of the factors of 12?

Solution:

The factors of 12 can be found as follows:

$1 \times 12 = 12$ or $12 \times 1 = 12$

$2 \times 6 = 12$ or $6 \times 2 = 12$

$3 \times 4 = 12$ or $4 \times 3 = 12$

Hence, the factors of 12 are $1, 2, 3, 4, 6, 12$.

Therefore, the sum of factors of

$12 = 1 + 2 + 3 + 4 + 6 + 12 = 28$.

Example:

What are the first five multiples of 7?

Solution:

Multiples of 7 through multiplication
$1 \times 7 = 7$
$2 \times 7 = 14$
$3 \times 7 = 21$
$4 \times 7 = 28$
$5 \times 7 = 35$

Therefore, the first 5 multiples of 7 are $7, 14, 21, 28$ and 35.

Example:

Find the first three common multiples of 2 and 14.

Solution:

Multiples of 2:

$2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 22, 24, 26, 28, 30,$

$32, 34, 36, 38, 40, 42, \ldots$

Multiples of 14 :

$14, 28, 42, 56, 70, 84, 98, \ldots$

Therefore, the first three common multiples of 2 and 14 are $14, 28$ and 42.

Example:

Which of the following is not a multiple of 23?

(A) 138 (B) 184

(C) 254 (D) 253

Solution:

Now, $23 \times 6 = 138$, $23 \times 8 = 184$ and $23 \times 11 = 253$, so clearly 254 is not a multiple of 23.

Note:

To understand what multiples are, let's just take an example of multiples of 3. The multiples are $3, 6, 9, 12, \ldots$. We find that every successive multiple appears as the third number after the previous one.

So if one wishes to find the number of multiples of 6 less than 255, we can arrive at the number through $\frac{255}{6} = 42$ (and the remainder 3). The remainder is of no consequence to us for this purpose. So in all there are 42 multiples.

If one wishes to find the multiples of 36 less than 255, find $\frac{255}{36} = 7$ (and the remainder is 3).

Hence, there are 7 multiples of 36.

Example:

Find the number of multiples of 7 which are less than 134 and greater than 45.

Solution: (13)

$\frac{134}{7} = 19 \quad$ (remainder 1)

There are 19 multiples of 7 which are less than 134.

$\frac{45}{7} = 6 \quad$ (remainder 3)

There are 6 multiples of 7 which are less than 45.

No.of multiples of 7 between 45 and 134

$$= 19 - 6 = 13$$

Factorization

It is the process of splitting any number into a form where it is expressed only in terms of the most basic prime factors.

For example, $36 = 2^2 \times 3^2$. It is expressed in the factorized form in terms of its basic prime factors.

The most commonly used prime factorization methods is the division method.

Follow the below steps to find the prime factorisation of a number using the division method:

- Step 1: Divide the given number by the smallest prime number that divides the number exactly.
- Step 2: Again, divide the quotient by the smallest prime number.
- Step 3: Repeat the process, until the quotient becomes 1.
- Step 4: Finally, multiply all the prime factors

Example:

Find the prime factorization of 460 using the division method.

Solution:

Step 1:

Start by dividing 460 by the smallest prime number, which is 2:

$460 \div 2 = 230$

Step 2:

230 is still even, so divide again by 2:

$230 \div 2 = 115$

Step 3:

115 is not divisible by 2, so try the next smallest prime number that divides 115, 5:

$115 \div 5 = 23$

Step 4:

23 is a prime number, so divide it by itself:

$23 \div 23 = 1$

Prime factorization of 460:

$460 = 2 \times 2 \times 5 \times 23$

Or

$460 = 2^2 \times 5 \times 23$

Example:

Express 108 as a product of its prime factors

(A) $2^2 \times 3^3$ (B) $2^3 \times 3^2$

(C) $2^1 \times 3^4$ (D) $2^4 \times 3^1$

Solution: (A)

By prime factorization, we get

2	108
2	54
3	27
3	9
	3

$108 = 2 \times 2 \times 3 \times 3 \times 3$

$= 2^2 \times 3^3$

Example:

If $3825 = 3^x \times 5^y \times 17^z$, then the value of $x+y-2z$ is

Solution: (2)

By prime factorization,

$3825 = 3^2 \times 5^2 \times 17$

Also given that, $3825 = 3^x \times 5^y \times 17^z$

$\Rightarrow x = 2, y = 2, z = 1$

$\therefore x + y - 2z = 2 + 2 - 2 \times 1$

$= 2$

Example:

Which of the following is not a prime factor of 3825?

(A) 3 (B) 5

(C) 11 (D) 17

Solution: (C)

By prime factorization we get,

$3825 = 3 \times 3 \times 5 \times 5 \times 17 = 3^2 \times 5^2 \times 17$

Hence, 11 is not a prime factor of 3825.

Example:

Find the smallest number by which 675 must be multiplied to obtain a perfect cube?

Solution:

Prime factorization of 675 : $3^3 \times 5^2$

To make it a perfect cube, each prime factor must appear in multiples of 3.

Here, 5 appears only twice, and it needs to appear three times.

$\Rightarrow$ The smallest number to multiply is 5

1. What is the least positive integer by which $2^5 \cdot 3^6 \cdot 4^3 \cdot 5^3 \cdot 6^7$ should be multiplied so that, the product is a perfect square?

[IOQM 2021]

2. The product $55 \times 60 \times 65$ is written as the product of five distinct positive integers. What is the least possible value of the largest of these integers?

[IOQM 2021]

3. Let $\overline{abc}$ be a three digit number with non-zero digits such that $a^2 + b^2 = c^2$. What is the largest possible prime factor of $\overline{abc}$?

[PRMO 2019]

4. Let n be the largest integer that is the product of exactly 3 distinct prime numbers, x, y, and $10x + y$, where x and y are digits. What is the sum of the digits of n ?

[PRMO 2015]

Number of Factors

For any composite number C, which can be expressed as

$C = a^p \times b^q \times c^r \times \ldots$

where $a, b, c \ldots$ are all distinct prime factors and p, q, r are positive integers, the number of factors is equal to

$(p+1) \times (q+1) \times (r+1) \times \ldots$

Sum of Factors

Sum of all factors of C

$$= \frac{(a^{p+1} - 1)}{a - 1} \times \frac{(b^{q+1} - 1)}{b - 1} \times \frac{(c^{r+1} - 1)}{c - 1} \times \ldots$$

Product of Factors

Product of factors of $C = C^{(\text{Total no. of factors}/2)}$

Example:

Find the total number of factors of 36 using its prime factorization.

Solution:

The prime factorization of 36 is given by:

$36 = 2 \times 2 \times 3 \times 3 = 2^2 \times 3^2$

Number of factors

$= (2+1) \times (2+1) = 3 \times 3 = 9$

Example:

How many total factors does 100 have?

Solution:

For $100 = 2^2 \times 5^2$, the exponents are 2 and 2.

So, total number of factors is:

$(2+1) \times (2+1) = 3 \times 3 = 9$

Example:

Find the sum of factors of 90

Solution:

$90 = 2 \times 3^2 \times 5$

Sum of factors

$= \dfrac{(2^{1+1} - 1)}{2-1} \times \dfrac{(3^{2+1} - 1)}{3-1} \times \dfrac{(5^{1+1} - 1)}{5-1}$

$= \dfrac{3}{1} \times \dfrac{26}{2} \times \dfrac{24}{4} = 3 \times 13 \times 6$

$= 234$

Example:

If $N = 12^3 \times 3^4 \times 5^2$, find the total number of even factors of N.

Solution:

The factorized form of N is

$(2^2 \times 3^1)^3 \times 3^4 \times 5^2 = 2^6 \times 3^7 \times 5^2$.

Hence, the total number of factors of N is $(6+1)(7+1)(2+1) = 7 \times 8 \times 3 = 168$.

Some of these are odd multiples and some are even. The odd multiples are formed only with the combination of $3s$ and $5s$.

So, the total number of odd factors is

$(7+1)(2+1) = 24$.

Therefore, the number of even factors is

$168 - 24 = 144$.

Example:

A number N when factorized can be written as $N = a^4 \times b^3 \times c^7$. Find the number of perfect squares which are factors of N (the three prime numbers $a, b, c > 2$).

Solution:

For a perfect square to divide N, the powers of its prime factors must be even.

The power of a can be 0, 2 or 4

(i.e., 3 possibilities)

The power of b can be 0 or 2

(i.e., 2 possibilities)

The power of c can be 0, 2, 4 or 6

(i.e., 4 possibilities)

Hence total number of such perfect squares is given by: $3 \times 2 \times 4 = 24$

Example:

Find the product of all the factors of 400

Solution:

$400 = 2^4 \times 5^2$

No. of factors $= (4+1)(2+1) = 5 \times 3 = 15$

Product of factors $= (400)^{15/2} = 20^{15}$

Example:

The product of all factors of a number N is equal to the number raised to the power four. If N is less than 100, then find the number of possible values of N?

Solution:

The product of all the factors of a natural number N with x factors is $N^{x/2}$.

As per the given condition

$N^{x/2} = N^4 = N^{8/2} \Rightarrow x = 8$

Therefore the number of factors of $N = 8$.

Now 8, can be expressed as 1×8 or 2×4 or $2 \times 2 \times 2$

$\Rightarrow N = a^7 (\text{for } x = 1 \times 8)$

or $N = a \times b^3 (\text{for } x = 2 \times 4)$

or $N = a \times b \times c (\text{for } x = 2 \times 2 \times 2)$

Where a, b, c are prime numbers.

Now the possible values of N less than 100 are:

$\Rightarrow (2^3 \times 3), (2^3 \times 5), (2^3 \times 7), (2^3 \times 11), (3^3 \times 2), (2 \times 3 \times 5), (2 \times 3 \times 7), (2 \times 3 \times 11), (2 \times 3 \times 13), (2 \times 5 \times 7)$

i.e. a total of 10 values.

5. How many two digit numbers have exactly 4 positive factors? (Here 1 and the number n are also considered as factors of n.)

[IOQM 2021]

6. A natural number k is such that $k^2 < 2014 < (k+1)^2$, what is the largest prime factor of k?

[PRMO 2014]

7. Let $S_n = n^2 + 20n + 12$, n a positive integer. What is the sum of all possible values of n for which S_n is a perfect square?

[PRMO 2012]

8. Let m be the smallest odd positive integer for which $1 + 2 + \ldots + m$ is a square of an integer and let n be the smallest even positive integer for which $1 + 2 + \ldots + n$ is a square of an integer. What is the value of $m + n$?

[PRMO 2013]

9. Let N denote the number of all natural numbers n such that n is divisible by a prime $p > \sqrt{n}$ and $p < 20$. What is the value of N?

[PRMO 2019]

10. What is the smallest prime number p such that $p^3 + 4p^2 + 4p$ has exactly 30 positive divisors?

[PRMO 2019]

HCF and LCM

- **LCM (Least Common Multiple):**

  The LCM of a given set of numbers, as the name suggests is the smallest positive number that is a multiple of each of the given numbers

- **HCF (Highest Common Factor):**

  The HCF of a given set of numbers, as the name suggests is the largest factor of the given set of numbers.

Consider the numbers 12, 20 and 30. The factors and the multiples are:

Factors	Given Numbers	Multiples
$1, 2, 3, 4, 6, 12$	12	$12, 24, 36, 48, 60, 72, 84, 96, 108, 120 \ldots\ldots$
$1, 2, 4, 5, 10, 20$	20	$20, 40, 60, 80, 100, 120 \ldots\ldots$
$1, 2, 3, 5, 6, 10, 15, 30$	30	$30, 60, 90, 120 \ldots\ldots$

The common factors are 1 and 2 and the common multiples are $60, 120 \ldots$

Thus, the HCF is 2 and the LCM is 60.

The meaning of HCF is the largest number that divides all the given numbers.

Also, since a number divides its multiple, the meaning of LCM is that it is the smallest number that can be divided by all the given numbers.

HCF will be lesser than or equal to the least of the numbers and LCM will be greater than or equal to the greatest of the numbers.

Example:

Find the HCF of 24 and 36.

Solution:

The factors of 24 are $1, 2, 3, 4, 6, 8, 12$ and 24.

The factors of 36 are $1, 2, 3, 4, 6, 9, 12, 18$, and 36.

Here, the greatest factor that is common in both the factors list is 12.

Therefore, the HCF of 24 and 36 is 12.

Example:

What is the LCM of 6 and 8?

Solution:

The multiples of 6 are $6, 12, 18, 24, 30, 36, \ldots$

The multiples of 8 are $8, 16, 24, 32, 40, 48, \ldots$

The smallest common multiple of 6 and 8 is 24.

Therefore, the LCM of 6 and 8 is 24.

How to Find LCM Using Prime Factorization

The given numbers are expressed as a product of its prime factors. We select the common prime factors with the highest exponent. Then, we multiply all such common factors with the prime factors that are not common. The product thus obtained is the LCM.

How to Find HCF Using Prime Factorization

The given numbers are expressed as a product of its prime factors. We select the common prime factors with the lowest exponent. Then, we multiply all such common factors. The result obtained is the HCF

Note:

For any two numbers x and y,

$x \times y = \text{HCF}(x, y) \times \text{LCM}(x, y)$

Example:

Using prime factorization, find HCF and LCM of 96 and 120.

Solution:

By prime factorization,

$96 = 2^5 \times 3^1$

$120 = 2^3 \times 3 \times 5$

$\therefore$ LCM $(96, 120) = 2^5 \times 3 \times 5 = 480$

HCF $(96, 120) = 2^3 \times 3 = 24$

Example:

Find the HCF and LCM of 12, 15 and 21, using the prime factorization method.

Solution:

By prime factorization

$12 = 2^2 \times 3$

$15 = 3 \times 5$

$21 = 3 \times 7$

$\text{LCM}(12, 15, 21) = 3 \times 5 \times 7 \times 2^2 = 420$

$\text{HCF}(12, 15, 21) = 3$

Example:

Two numbers are in the ratio 17 : 13. If their HCF is 15. What are the numbers?

Solution:

Let the two numbers be represented as:

(i) First number, $a = 17x$

(ii) Second number, $b = 13x$

Here, x is a common multiplier.

Since the HCF of the two numbers a and b is equal to x, we can set $x = 15$

Substitute x back into the expressions for a and b:

First number: $a = 17 \times 15 = 255$

Second number: $b = 13 \times 15 = 195$

Therefore, the two numbers based on the given ratio and HCF are 255 and 195

Example:

If the $\text{HCF}(2520, 6600) = 40$ and $\text{LCM}(2520, 6600) = 252 \times k$, then the value of k is

Solution:

We have, $\text{HCF}(a, b) \times \text{LCM}(a, b) = a \times b$

Here, $\text{HCF}(2520, 6600) \times \text{LCM}(2520, 6600)$

$$= 2520 \times 6600$$

i.e, $40 \times 252 \times k = 2520 \times 6600$

So, $k = 1650$

Example:

LCM of two distinct natural numbers is 211. What is their HCF?

Solution:

211 is a prime number. So there is only one pair of distinct numbers possible whose LCM is 211, i.e., 1 and 211.

HCF of 1 and 211 is 1.

Example:

An army contingent of 616 members is to march behind an army band of 48 members in a parade. The two groups are to march in the same number of columns. What is the maximum number of columns in which they can march?

Solution:

Clearly, the maximum number of columns

$= \text{HCF}(616, 48)$

Now, $616 = 2^3 \times 7 \times 11$ and $48 = 2^4 \times 3$

$\therefore$ $\text{HCF}(616, 48) = 2^3 = 8$

$\therefore$ Required number of columns $= 8$

Example:

4 bells toll together at 9.00 a.m. They toll after $7, 8, 11$ and 12 seconds respectively. How many times will they toll together again in the next 3 hours?

Solution:

We need to find out the LCM (Least Common Multiple) of $7, 8, 11$ and 12 to determine the time taken for all four bells to toll together again.

Calculating LCM:

$7 = 7 \times 1$

$8 = 2 \times 2 \times 2$

$11 = 11 \times 1$

$12 = 2 \times 2 \times 3$

LCM of $7, 8, 11$ and

$12 = 2^3 \times 3 \times 7 \times 11 = 1848$ seconds.

This means all four bells will toll together again after 1848 seconds or 30 minutes and 48 seconds.

Now, we need to find out how many times all four bells will toll together in the next 3 hours, i.e., 180 minutes.

Calculating number of times:

Number of times in 1 hour $= 60$ minutes$/30.8$ minutes (time taken by all four bells to toll together) $= 1.95$

Number of times in 3 hours $= 3 \times 1.95 = 5.85$

As we cannot have fractional tolls, we need to take the floor value which is 5.

Example:

In a teacher's workshop, the number of teachers teaching French, Hindi and English are 48, 80 and 144 respectively. Find the minimum number of rooms required if in each room the same number of teachers are seated and all of them are of the same subject.

Solution:

The number of rooms will be minimum if each room accommodates a maximum number of teachers. Since in each room, the same number of teachers are to be seated and all of them must be of the same subject, the number of teachers in each room must be HCF of $48, 80$ and 144

By prime factorization,

$48 = 2^4 \times 3, 80 = 2^4 \times 5$ and $144 = 2^4 \times 3^2$

$\therefore$ HCF$(48, 80, 144) = 2^4 = 16$

$\therefore$ The maximum number of teachers per room is 16

The number of rooms required for teachers teaching French $= \dfrac{48}{16} = 3$

The number of rooms required for teachers teaching Hindi $= \dfrac{80}{16} = 5$

The number of rooms required for teachers teaching English $= \dfrac{144}{16} = 9$

Then, the total number of rooms required

$= 3 + 5 + 9 = 17$

Hence, the minimum number of rooms required $= 17$

Note:

HCF$(p,q,r)\times$ LCM$(p,q,r) \neq p \times q \times r$, where p,q,r are positive integers. However, the following results hold good for three numbers p,q and r:

$$\text{LCM}(p,q,r) = \frac{p \cdot q \cdot r \cdot \text{HCF}(p,q,r)}{\text{HCF}(p,q) \cdot \text{HCF}(q,r) \cdot \text{HCF}(p,r)}$$

$$\text{HCF}(p,q,r) = \frac{p \cdot q \cdot r \cdot \text{LCM}(p,q,r)}{\text{LCM}(p,q) \cdot \text{LCM}(q,r) \cdot \text{LCM}(p,r)}$$

Some Important Points

Note:

If we have to find the greatest number that will exactly divide p,q and r then the required number is HCF of p,q,r.

Example:

Find the greatest number that will exactly divide 65, 52 and 78.

Solution:

Required number $=$ HCF$(65, 52, 78) = 13$

Note:

If we have to find the greatest number that will divide p,q and r leaving remainders a,b and c respectively, then the required number is HCF of $(p-a),(q-b)$ and $(r-c)$.

Example:

Find the greatest number that will divide 65, 52 and 78 leaving remainders 5, 2 and 8 respectively.

Solution:

Required number

$= \text{HCF of } (65-5), (52-2), (78-8) = 10$

Note:

If we have to find the greatest number that will divide p, q and r leaving the same remainder in each case then the required number is HCF of the absolute values of $(p-q), (q-r)$ and $(r-p)$.

Example:

Find the greatest number that will divide 65, 81 and 145 leaving the same remainder in each case.

Solution:

$$\begin{aligned}\text{Required number} &= \text{HCF of } (81-65), \\ &(145-81) \text{ and } (145-65) \\ &= \text{HCF of } 16, 64, 80 = 16\end{aligned}$$

Note:

If we have to find the least number which is exactly divisible by p, q and r then the required number is LCM of $p, q,$ and r.

Example:

Find the least number that is exactly divisible by 6, 5 and 7.

Solution

Required number $=$ LCM of 6, 5 and 7 $= 210$.

Note:

If we have to find the least number which when divided by p, q and r leaves the remainder a, b and c respectively and it is observed that $(p-a) = (q-b) = (r-c) = K$ (say) then the required number $= \text{LCM}(p, q, r) - K$

Example:

Find the least number which, when divided by 6, 7 and 9 leaves the remainders 1, 2 and 4 respectively.

Solution:

Here $(6-1) = (7-2) = (9-4) = 5$

$$\begin{aligned}\text{So the required number } &= \text{LCM}(6, 7, 9) - 5 \\ &= 126 - 5 = 121\end{aligned}$$

Note:

If we have to find the least number which when divided by p, q and r leaves the same remainder 'a' in each case, then the required number is $\text{LCM}(p, q, r) + a$

Example:

Find the least number which when divided by 15, 20 and 30 leaves the remainder 5 in each case.

Solution:

$$\begin{aligned}\text{Required number } &= \text{LCM}(15, 20, 30) + 5 \\ &= 60 + 5 = 65\end{aligned}$$

11. When a three digit number is divided by $2, 3, 4, 5$, and 7, the remainders are all 1. Find the minimum and maximum values of such three digit numbers.

12. How many ordered pairs (a, b) of positive integers with $a < b$ and $100 \leq a, b \leq 1000$ satisfy $\gcd(a, b) : \operatorname{lcm}(a, b) = 1 : 495$?

[PRMO 2019]

13. Find all pairs of positive integer (x, y) such that HCF $(x, y) +$ LCM$(x, y) = 91$.

HCF and LCM of Fractions

HCF and LCM of fractions:

$$\text{LCM of fractions} = \frac{\text{LCM of numerators}}{\text{HCF of denominators}}$$

$$\text{HCF of fractions} = \frac{\text{HCF of numerators}}{\text{LCM of denominators}}$$

Make sure the fractions are in the most reduced form.

Example:

The LCM of the fractions $\frac{2}{3}$ and $\frac{5}{6}$ is

(A) $\frac{5}{9}$ (B) $\frac{10}{3}$

(C) $\frac{7}{6}$ (D) $\frac{5}{6}$

Solution:

Numerator: LCM of 2 and $5 = 10$

Denominator: HCF of 3 and $6 = 3$

$$\text{LCM of fractions} = \frac{\text{LCM of numerators}}{\text{HCF of denominators}} = \frac{10}{3}$$

Example:

The HCF of the fractions $\frac{4}{9}$ and $\frac{6}{15}$ is

(A) $\frac{2}{45}$

(B) $\frac{4}{15}$

(C) $\frac{6}{45}$

(D) $\frac{1}{9}$

Solution:

Numerator: HCF of 4 and 6 $= 2$

Denominator: LCM of 9 and 15 $= 45$

$$\text{HCF of fractions} = \frac{\text{HCF of numerators}}{\text{LCM of denominators}} = \frac{2}{45}$$

Example:

Find the HCF and LCM of $\frac{8}{9}, \frac{16}{81}, \frac{2}{3}$ and $\frac{10}{27}$

Solution:

HCF of given fractions

$$= \frac{\text{HCF of } (2, 8, 16, 10)}{\text{LCM of } (3, 9, 81, 27)} = \frac{2}{81}$$

LCM of given fractions

$$= \frac{\text{LCM of } (2, 8, 16, 10)}{\text{HCF of } (3, 9, 81, 27)} = \frac{80}{3}.$$

Division Algorithm

General representation of result is,

$$\frac{\text{Dividend}}{\text{Divisor}} = \text{Quotient} + \frac{\text{Remainder}}{\text{Divisor}}$$

Dividend $=$ (Divisor $\times$ Quotient)$+$ Remainder

Example:

What is the remainder obtained when 235 is divided by 7?

Solution:

We can write, $235 = 33 \times 7 + 4$

33 is the quotient when 235 is divided by 7 and 4 is the remainder.

Example:

The least number that must be added to 3105 to get a number exactly divisible by 21.

Solution

$3105 = 147 \times 21 + 18$

Remainder is 18. So $3(= 21 - 18)$, is the least number that must be added to 3105 to make it exactly divisible by 21.

Example:

A number being successively divided by 3, 5 and 8 leaves remainders 1, 4 and 7, respectively. Find the respective remainders if the order of divisors are reversed.

Solution:

3	x	
5	y	1
8	z	4
	1	7

$\therefore z = (8 \times 1 + 7) = 15;$

$y = (5z + 4) = (5 \times 15 + 4) = 79;$

$x = (3y + 1) = (3 \times 79 + 1) = 238$

Now

8	238	
5	29	6
3	5	4
	1	2

$\therefore$ Respective remainders are $6, 4, 2$.

14. A number when divided by 259 leaves a remainder 139. What will be the remainder when the same number is divided by 37?

Divisibility Rules

Divisibility Rule for $2, 4, 8$ and 16

- A number is divisible by 2 if the last digit of the number is divisible by 2 or 2^1.
- A number is divisible by 4 if the last two digit of the number is divisible by 4 or 2^2.
- A number is divisible by 8 if the last three digits of the number is divisible by 8 or 2^3.
- A number is divisible by 16 if the last four digits of the number is divisible by 16 or 2^4.

Example:

Check whether the number 2024 is divisible by 4.

Solution:

In the given number 2024, the last 2 digits are 24. 24 is divisible by 4. Therefore, 2024 is divisible by 4.

Example:

Is the number 2048 divisible by 8?

Solution:

The last three digits of 2048 are 048, which is 48.

$48 \div 8 = 6$ (no remainder).

Therefore, 2048 is divisible by 8.

Example:

Is the number 32768 divisible by 16?

Solution:

The last four digits of 32768 are 2768.

$2768 \div 16 = 173$ (no remainder).

Therefore, 32768 is divisible by 16.

Divisibility Rule for 5, 25, 125

- ❖ A number is divisible by 5 if the last digit of the number is divisible by 5 or 5^1.
- ❖ A number is divisible by 25 if last two digits of the number form a number that is divisible by 25 or 5^2.
- ❖ A number is divisible by 125 if the last three digits of the number form a number that is divisible by 125 or 5^3.

Example:

Is the number 3175 divisible by 25?

Solution:

Last two digits of 3175 are 75.

$75 \div 25 = 3$ (no remainder).

Since 75 is divisible by 25,3175 is divisible by 25.

Example:

Is the number 874750 divisible by 125?

Solution:

Last three digits of 874750 are 750.

$750 \div 125 = 6$ (no remainder).

Since 750 is divisible by 125, 874750 is divisible by 125.

Divisibility Rule for 3 and 9

- ❖ A number is divisible by 3 if the sum of the digits of the number is divisible by 3.
- ❖ A number is divisible by 9 if the sum of the digits of the number is divisible by 9.

Example:

Is the number 537 divisible by 3?

Solution:

Sum of the digits $= 5 + 3 + 7 = 15$

$15 \div 3 = 5$ (no remainder).

So, 537 is divisible by 3.

Example:

Is the number 4689 divisible by 3 and/or 9?

Solution:

Sum of the digits $= 4 + 6 + 8 + 9 = 27$

$27 \div 3 = 9$ (no remainder) $\rightarrow$ Divisible by 3

$27 \div 9 = 3$ (no remainder) $\rightarrow$ Divisible by 9

So, 4689 is divisible by both 3 and 9.

Divisibility Rule for 7, 11 and 13

- If you double the last digit and subtract it from the rest of the number and the answer is divisible by 7, the number is divisible by 7. You can apply this rule to that answer again if necessary.
- To find out if a number is divisible by 11, add every even-place digit, and call that sum 'x'. Add together the remaining digits, and call that sum 'y'. Take the positive difference of x and y. If the difference is zero or a multiple of eleven, then number is divisible by 11 . Repeat the rule if necessary.
- Delete the last digit from the number and then subtract 9 times the deleted digit from the remaining number. If what is left is divisible by 13, then number is divisible by 13. Repeat the rule if necessary.
- If the positive difference of the last three digit and the rest of the digits is divisible by 7, 11 or 13, then the number is divisibly by 7, 11 or 13, respectively.

Example:

Check whether 413 is divisible by 7 or not.

Solution:

Last digit $= 3$, remaining number $= 41$,

$41 - (3 \times 2) = 35$ (divisible by 7),

i.e., 413 is divisible by 7.

This rule can also be used for numbers having more than 3 digits.

Example:

Check whether 6545 is divisible by 7 or not.

Solution:

Last digit $= 5$, remaining number 654,

$654 - (5 \times 2) = 644$;

$64 - (4 \times 2) = 56$ divisible by 7,

i.e., 6545 is divisible by 7.

Example:

Is 2848 divisible by 11?

Solution:

Sum of even-place digits:

$2 + 4 = 6$

Sum of odd-place digits:

$8 + 8 = 16$

The difference between 6 and

$16 = 16 - 6 = 10$, which is not divisible by 11.

Therefore, 2848 is not divisible by 11 .

Example:

Check whether 234 is divisible by 13 or not.

Solution:

$(4 \times 9) - 23 = 13$ (divisible by 13),

i.e., 234 is divisible by 13.

Divisibility Rule for 17 and 19

- For 17: Take five times the last digit of the number and subtract it from the rest of the number. The result obtained should be either 0 or divisible by 17.
- For 19: Double the last digit of given number and add to remaining number. The result obtained should be divisible by 19.

Example:

Check whether 357 is divisible by 17 or not.

Solution:

$(7 \times 5) - 35 = 0$, $\Rightarrow$ 357 is divisible by 17.

Example:

Check whether 589 is divisible by 19 or not.

Solution:

$(9 \times 2) + 58 = 76$ (divisible by 19),

i.e., the number is divisible by 19.

Divisibility Rule for 6, 10, 12, 14, 15, 18, 24 and 36

- A number is divisible by 6, if the number is divisible by both 2 and 3.
- A number is divisible by 10, if the number is divisible by both 2 and 5.
- A number is divisible by 12, if the number is divisible by both 3 and 4.
- A number is divisible by 14, if the number is divisible by both 2 and 7.
- A number is divisible by 15, if the number is divisible by both 3 and 5.
- A number is divisible by 18, if the number is divisible by both 2 and 9.
- A number is divisible by 24, if the number is divisible by both 3 and 8.
- A number is divisible by 36, if the number is divisible by both 4 and 9.

Example:

Check whether 1440 is divisible by 15.

Solution:

A number is divisible by 15 if it is divisible by both 3 and 5.

Since the unit digit of 1440 is 0, it is divisible by 5.

Also, the sum of digits of

$1440 = 1 + 4 + 4 + 0 = 9$

Hence it is divisible by 3.

Since 1440 is divisible by both 3 and 5, 1440 is divisible by 15.

Example:

Is the number 246 divisible by 6?

Solution:

A number is divisible by 6 if it is divisible by both 2 and 3.

Since 246 ends with 6 (an even digit), it is divisible by 2.

The sum of the digits is $2 + 4 + 6 = 12$, which is divisible by 3.

Since 246 is divisible by both 2 and 3, it is divisible by 6.

Example:

Is the number 36,432 divisible by 12?

Solution:

A number is divisible by 12 if it is divisible by both 3 and 4.

The sum of the digits is

$3 + 6 + 4 + 3 + 2 = 18$, which is divisible by 3.

Since the last two digits form the number 32, and 32 is divisible by 4, the number is divisible by 4.

As 36,432 is divisible by both 3 and 4, it is divisible by 12 .

Example:

Is the number 5184 divisible by 24?

Solution:

A number is divisible by 24 if it is divisible by both 8 and 3.

The sum of the digits is $5 + 1 + 8 + 4 = 18$, which is divisible by 3.

The last three digits are 184, and since 184 is divisible by 8 (because $8 \times 23 = 184$), the number is divisible by 8.

As 5,184 is divisible by both 3 and 8, it is divisible by 24.

Example:

Is the number 1,372 divisible by 14?

Solution:

A number is divisible by 14 if it is divisible by both 7 and 2.

Since 1,372 ends with 2, which is even, it is divisible by 2.

To check divisibility by 7, double the last digit $(2 \times 2 = 4)$ and subtract it from the remaining part of the number $(137 - 4 = 133)$.

Now, check if 133 is divisible by 7. Repeat the process: double the last digit $(3 \times 2 = 6)$ and subtract from the rest $(13 - 6 = 7)$.

Since 7 is divisible by 7, the original number 1,372 is divisible by 7.

Because 1,372 is divisible by both 2 and 7, it is divisible by 14.

Example:

For what digit(s) x will a 7-digit number $3xx6xx2$ be divisible by 4?

Solution:

$1, 3, 5, 7$ and 9

The two-digit number $x2$ needs to be divisible by 4 by the divisibility rule

The following numbers work: $12, 32, 52, 72$ and 92

Example:

Find the largest four digit number that when reduced by 54, is perfectly divisible by all even natural numbers less than 20.

Solution:

Even natural numbers less than 20 are $2, 4, 6, 8, 10, 12, 14, 16, 18$.

Their LCM $= 2\times$ LCM of first 9 natural numbers $= 2\times\ 2520 = 5040$.

This happens to be the largest four-digit number divisible by all even natural numbers less than 20.

54 was subtracted from our required number to get this number.

Hence, required number $= 5040 + 54 = 5094$

15. Which digits should come in place of * and \$ if the number $62684*\$$ is divisible by both 8 and 5?

16. How many five-digit number multiples of 11 are there, if the five digit are $7, 6, 5, 4$, and 3 in some order?

Euclid's Division Lemma

If we have two positive integers a and b, then there exist unique integer q and r, which satisfies the condition $a = b \times q + r$ where $0 \leq r < b$.

Euclid's Division Algorithm

To obtain the HCF of two positive integers, say c and d, with $c > d$, follow the steps below:
Step 1: Apply Euclid's Division Lemma to c and d. So, we find whole numbers, q and r such that $c = dq + r,\ 0 \leq r < d$
Step 2: If $r = 0$, d is the HCF of c and d. If $r \neq 0$, apply the division lemma to d and r.
Step 3: Continue the process till the remainder is zero. The divisor at this stage will be the required HCF
This algorithm works because
HCF(c, d) = HCF(d, r) where the symbol HCF(c, d) denotes the HCF of c and d, etc.

Euclid's Division Algorithm is a technique to compute the HCF of two given positive integers. Recall that the HCF of two positive integers a and b is the largest positive integer d that divides both a and b.
Let us see how the algorithm works, through an example first. Suppose we need to find the HCF of the integers 455 and 42. We start with the larger integer, that is, 455. Then we use Euclid's lemma to get
$455 = 42 \times 10 + 35$
Now consider the divisor 42 and the remainder 35, and apply the division lemma to get
$42 = 35 \times 1 + 7$
Now consider the divisor 35 and the remainder 7, and apply the division lemma to get
$35 = 7 \times 5 + 0$
Notice that the remainder has become zero, and we cannot proceed any further.
HCF of 455 and 42 is the divisor at this stage, i.e., 7.

Example:

Use Euclid's algorithm to find the HCF of 4052 and 12576.

Solution:

Step 1: Since $12576 > 4052$, we apply the division lemma to 12576 and 4052, to get

$12576 = 4052 \times 3 + 420$

Step 2: Since the remainder $420 \neq 0$, we apply the division lemma to 4052 and 420, to get

$4052 = 420 \times 9 + 272$

Step 3: We consider the new divisor 420 and the new remainder 272, and apply the division lemma to get

$420 = 272 \times 1 + 148$

We consider the new divisor 272 and the new remainder 148, and apply the division lemma to get

$272 = 148 \times 1 + 124$

We consider the new divisor 148 and the new remainder 124, and apply the division lemma to get

$148 = 124 \times 1 + 24$

We consider the new divisor 124 and the new remainder 24, and apply the division lemma to get

$124 = 24 \times 5 + 4$

We consider the new divisor 24 and the new remainder 4, and apply the division lemma to get

$24 = 4 \times 6 + 0$

The remainder has now become zero, so our procedure stops. Since the divisor at this stage is 4, the HCF of 12576 and 4052 is 4.

Notice that

$4 = \text{HCF}(24, 4) = \text{HCF}(124, 24)$

$= \text{HCF}(148, 124) = \text{HCF}(272, 148)$

$= \text{HCF}(420, 272) = \text{HCF}(4052, 420)$

$= \text{HCF}(12576, 4052)$

Example:

Find HCF of 3675 and 42 using Euclid's division algorithm.

Solution:

Step 1: $3675 = 42 \times 87 + 21$

Remainder is not zero. Now 42 becomes the new dividend and 21 becomes the new divisor.

Step 2: $42 = 21 \times 2 + 0$

Remainder becomes zero here

$\therefore$ the divisor at this step is the HCF of 3675 and 42

$\therefore$ HCF$(3675, 42) = 21$.

17. Two numbers, x and y, are such that when divided by 6, they leave remainders 4 and 5, respectively.

Find the remainder when $(x^2 + y^2)$ is divided by 6.

(1) 15	(5) 30	(9) 69	(13) 8	(16) 12
(2) 20	(6) 11	(10) 43	(14) 28	(17) 5
(3) 29	(7) 6	(11) 421, 841	(15) $* \to 4$ or 8, $\$ \to 0$	
(4) 12	(8) 9	(12) 20		

Space for rough work